

Referee report on
“Back to the 1980s or Not? The Drivers of Inflation and Real Risks in Treasury Bonds” (Manuscript 2023-1106)

Summary.

The author investigates how aggregate demand, supply and monetary policy drive bond-stock betas and argues that bond-stock betas can be interpreted as forward looking indicators of economic outcomes. In the 1980s, nominal bond betas were notably higher than inflation-indexed bond betas, reflecting volatile supply-driven inflation and an inflation-focused monetary policy. As inflation became more demand-driven from 2000 and monetary policy became output-focused, nominal bond betas changed sign and the gap between nominal and inflation-indexed bond narrowed. Interestingly, the author suggests the post-pandemic era reveals a distinct equilibrium in which supply-driven inflation gets combined with an inertial and output-focused monetary policy.

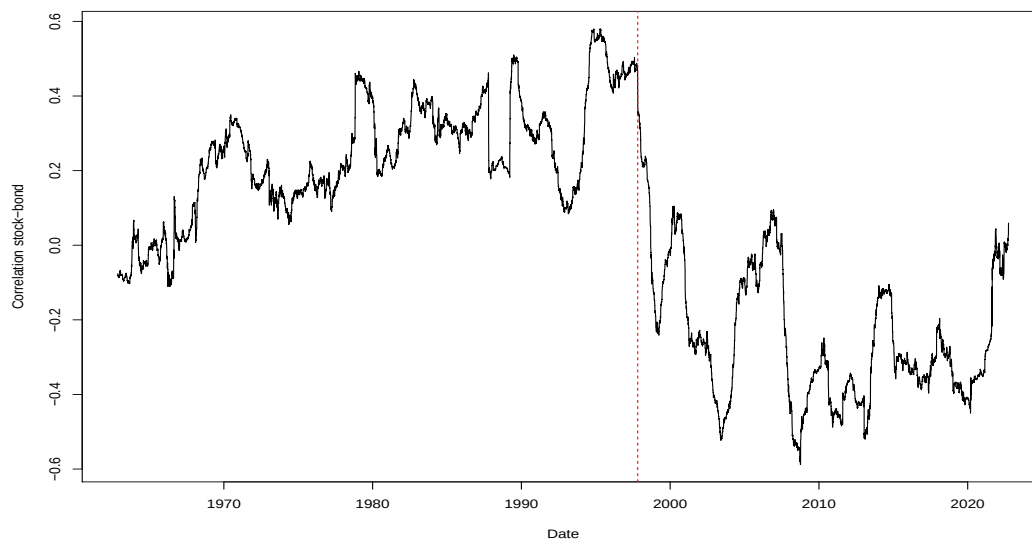
I believe the paper is interesting and the question is ambitious. However, I don't think that this paper has made a sufficient contribution to the literature to clear the very high bar at the *American Economic Review*. Below I list some observations, comments, and suggestions for improvement.

Comments.

1. Figure 1 is a key figure in the paper. It provides the motivation to study the above-mentioned third equilibrium. Yet, I don't really see a structural break in Figure 1 and there is no formal test. By looking at Panel A, the nominal-bond and inflation-indexed bond betas appear remarkably stable from 2009 to the present. I don't believe there is anything special in that picture about the post-pandemic period. Estimating the optimal number of breaks and their location, showing robustness of these estimates and having better estimates of betas are all necessary steps to convince the reader that there exists a “new” equilibrium.
2. If we assume that a new post-pandemic equilibrium indeed exists, it seems overly simplistic to assert that the model can only generate it by starting with the calibration for the 1980s and combining it with an output-focused monetary policy rule. The author should either re-estimate the model for the post-pandemic period or demonstrate that the described combination is the only method to obtain the new equilibrium. Currently, this is vaguely claimed in Section 4.1. Why is the calibration of the first subsample the right starting point? If one believes consumers accumulated savings during the

pandemics and then spent everything at once, can this imply a different distribution of supply/demand shocks compared to the one in the 80s?

3. As the author states, “the bond preference shock is the only shock in the model that induces a negative correlation between real bond prices and the output gap” and it is necessary to explain negative real bond-stock betas during the pre-pandemic 2000s. I understand the author has tried to microfound this shock in the Appendix, but the question remains: What is this shock? Even if the author says “My results do not depend on the specific interpretation of ξ_t within these broad categories,” this cannot be true. Quantitatively the two explanations must give different results: the treasury convenience or intermediary frictions in the first class of microfoundations cannot have the same exact time variation as investors’ optimism. If we believe that ξ is captured by a credit spread shock á la Gilchrist and Zakrajsek (2012), as it is stated in the introduction, then why don’t we use directly the time-series of the GZ credit spread in the model?
4. There is no effort to discriminate among competing models and alternative hypotheses. Are the facts the model matches unique to this model? Despite what may appear from my previous comment, I like the preference shock, but as I said above we need to understand what it captures. A line of argument relating ξ to treasury convenience could be as follows. The correlation between stock and bond returns changed virtually in one day, when the Asian crisis caused risk contagion across global markets on 10/27/1997. The plot below reports the correlation between the value-weighted market raw returns (from Kenneth French’s website) and the returns on the 5-year (zero coupon) nominal Treasury bonds from June 1961 to September 2022 at daily frequency. Correlation is estimated based on the past 360 daily returns. The vertical line represents 10/27/1997.



Investors across the globe sold risky assets, and bought US Treasury bonds, driving yields down. Such a global “risk-off” event resulted in one of the worst one-day returns for the Nasdaq and Dow Jones, and Treasury yields plummeted to two-year lows. In the following year, to respond to the crisis, the Federal Reserve first put off a rate increase, and subsequently decided to reduce rates three times, reinforcing the positive correlation between equities and yields. Similar events occurred twice

more in the year after with the LTCM and Russia debt crises. These events established treasury bonds as the *safe asset of choice*. It is instructive that another period in which the negative correlation between stock and bond returns got stronger was the 2008 financial crisis.

Such a precautionary savings behavior, and the increasing *conveniency* of Treasury bonds, could lie behind the change in correlation between stock and bond returns. Moreover, He, Nagel and Song (2022) discuss the treasury *inconvenience* during March 2020. I believe this manuscript and the work by He, Nagel and Song (2022) are closely related. However, their argument is quite different and seems more convincing: “the Treasury market experienced severe stress and illiquidity during the COVID-19 crisis, raising concerns that the safe-haven status of US Treasuries may be eroding.” They also “document large shifts in Treasury ownership and temporary accumulation of Treasury and reverse repo positions on dealer balance sheets during this period.” How temporary is the effect He, Nagel and Song document, and can the author quantify the role of a shift in ownership/preferences vs the current explanation?

5. Investors need to learn about potential policy changes. Moreover, when forming beliefs, they may consider a wide range of potential policies. How investors form beliefs about future regimes plays a central role that in this manuscript is just ignored. How abrupt was the monetary policy regime change?
6. Can the author find direct evidence of a change in the monetary policy rule consistent with the one implied by the paper? Shapiro and Wilson (2022) estimated a time-varying inflation target from textual analysis showing it went down from 3% in the 80s to 1.37. Can the model accommodate this pattern? What would happen if, rather than a radical change in the monetary policy rule, the Fed’s inflation target had gradually moved down?
7. What should have the Fed done in the post-pandemic period? Reading the paper, I kept on wondering whether the model implies that the response to inflation was too lax. Can the paper say anything about optimal monetary policy response?

Minor comments

1. I am not sure what the right interpretation of the results in Section 4.4 should be. If the main conclusion is that bond-stock betas reflect investors’ views of the data generating process (despite the realized shocks) this seems obvious and the same argument can be made in simpler ways.
2. The fact that γ^x reaches the upper bound suggests that the model is not properly estimated, despite the economic priors. Moreover, assume γ^x did not reach the upper bound. Could the author discuss local vs global minima?
3. How does the author select among multiple equilibria?